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BSc (Hons) Literature Review Guide and Final Project Differences

Software Engineering Final Development Project vs Data Science Final Research Development Project

Prepared for	Academic project guidance / evaluation support
Programme focus	BSc (Hons) Software Engineering and BSc (Hons) Data Science
Main purpose	To clarify literature review expectations and final project differences
Language	English

This guide can be used by students, supervisors, and evaluators to understand how the literature review should be written differently for software-oriented and data-oriented final projects, especially when AI integration is included.

1. Introduction

A literature review in a final-year project is not simply a summary of articles. It is an academic discussion that explains what is already known, what methods have been used, what limitations remain, and how the proposed project addresses a clear gap. However, the focus of the literature review changes depending on the degree programme and the nature of the final project.

For a BSc (Hons) in Software Engineering, the final project is normally a Final Development Project. The main outcome is a working software system. If AI is included, the AI component is usually integrated into the system as a feature, service, or intelligent module.

For a BSc (Hons) in Data Science, the final project is normally a Final Research Development Project. The main outcome is not only a working system but also a research-based data science solution, including dataset selection, preprocessing, modelling, evaluation, interpretation, and research findings.

2. Literature Review Requirements for a BSc (Hons) Data Science Research Development Project

A Data Science literature review should strongly connect the research problem, data, algorithms, experiments, and evaluation metrics. It must show that the student understands both the domain problem and the scientific methods used to solve it.

Component	What to Include
Introduction to the Literature Review	Briefly introduce the research area, problem domain, purpose of the review, and how the chapter supports the proposed project.
Research Domain and Problem Background	Explain the real-world problem, why it is important, who is affected, and why data science is suitable for solving or analysing the problem.
Review of Previous Studies	Review at least 10-15 relevant academic sources such as journal papers, conference papers, and high-quality technical reports. Discuss authors, year, aims, methods, datasets, results, strengths, and limitations.
Dataset Review	Discuss datasets used in previous studies, dataset sources, variables/features, target variable, sample size, missing values, imbalance, bias, data quality, and relevance to the proposed project.
Data Preprocessing and Feature Engineering	Explain literature related to cleaning, encoding, scaling, normalization, outlier handling, missing value treatment, dimensionality reduction, feature selection, and data balancing.
Machine Learning / Deep Learning Models	Discuss algorithms relevant to the project, such as Logistic Regression, Random Forest, SVM, XGBoost, CNN, LSTM, Transformers, clustering models, or time-series models.
Model Evaluation Metrics	Explain suitable evaluation metrics such as accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, MAE, RMSE, R2 score, silhouette score, or other domain-specific measures.
Tools and Technologies	Discuss why Python, Jupyter/Colab, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, SQL, Power BI, Tableau, Streamlit, Flask, or other tools are suitable.
Ethical, Privacy, Bias, and Security Issues	Discuss responsible AI, data privacy, consent, bias, fairness, explainability, transparency, and limitations of public datasets.
Critical Analysis	Compare studies critically. Do not only describe them. Identify weaknesses such as small datasets, lack of validation, poor explainability, low generalisability, and limited real-world testing.
Research Gap	Clearly state what previous research has not solved and how the proposed project will address that gap through data, methods, modelling, evaluation, or implementation.
Conceptual / Theoretical Framework	Present a high-level flow such as data collection -> preprocessing -> feature engineering -> model training -> evaluation -> prediction -> report or dashboard output.

Typical Data Science Project Flow

Data Collection	**Preprocessing**	**Feature Engineering**	**Model Training**	**Evaluation**	**Prediction / Output**

3. Main Difference Between the Two Final Projects

The most important difference is the centre of gravity of the project. In Software Engineering, the centre is the software system. In Data Science, the centre is the data-driven research problem and the model evidence.

Area	BSc (Hons) Software Engineering	BSc (Hons) Data Science
Project nature	Final Development Project	Final Research Development Project
Primary aim	Design, develop, test, and deploy a functional software system.	Investigate a research problem using data, models, experiments, and evidence.
Main output	A working system, application, web platform, mobile app, or software solution.	A data science model or pipeline, research findings, evaluated results, and possibly a prototype/system.
Role of AI	AI is usually an integrated feature such as chatbot, recommendation, prediction, automation, image recognition, or API/module integration.	AI/ML is usually the core research method. The student must justify dataset, model choice, training, testing, and evaluation.
Success measure	System completeness, functional requirements, usability, software quality, testing, maintainability, and deployment readiness.	Model performance, data quality, experimental design, validity of findings, evaluation metrics, interpretation, and research contribution.
Novelty expectation	Practical software solution, improved workflow, better user experience, integration, automation, or system innovation.	Research gap, better model performance, meaningful comparison, new feature set, improved method, or validated data-driven insight.

4. Differences in the Literature Review

Both degrees require academic references and critical analysis. However, the literature review must support different project outcomes. The following table shows how the focus changes.

Literature Review Area	Software Engineering Final Development Project	Data Science Final Research Development Project
Purpose of literature review	To justify the need for the software system and support requirements, design, architecture, technologies, and testing approaches.	To justify the research problem, dataset, data science methods, algorithms, evaluation metrics, and research gap.
Existing work review	Review existing systems, applications, platforms, software solutions, frameworks, and user problems.	Review previous research studies, datasets, algorithms, experimental methods, model results, and limitations.
Problem analysis	Focuses on user needs, business process problems, system inefficiencies, manual processes, usability issues, and technical limitations.	Focuses on data-driven problem definition, prediction/classification/forecasting/analysis objectives, research questions, and measurable outcomes.
Technical literature	Software architecture, design patterns, SDLC models, requirement engineering, database design, API integration, UI/UX, security, cloud deployment, and testing.	Statistics, machine learning, deep learning, data preprocessing, feature engineering, model selection, hyperparameter tuning, validation, explainability, and bias.
AI integration discussion	Should explain how AI improves the system as a feature and how it integrates with the software architecture. Example: AI (Own Data Trained) chatbot integrated into a student support system.	Should explain AI/ML as the central method. Example: compare Random Forest, XGBoost, and Neural Networks using a selected dataset and evaluation metrics. AI (Own Data Trained) chatbot integrated into a student support system.
Dataset importance	Dataset may be used to support an AI feature, testing, sample data, or database content, but the software system remains the main artefact.	Dataset is central. Literature must discuss data sources, variables, preprocessing, quality, bias, imbalance, and suitability.
Evaluation literature	Focuses on software testing methods: unit testing, integration testing, system testing, user acceptance testing, usability testing, performance testing, and security testing.	Focuses on model evaluation: train-test split, cross-validation, accuracy, precision, recall, F1-score, AUC, RMSE, MAE, confusion matrix, statistical validity, and model comparison.
Critical analysis	Critically evaluates limitations of existing systems: lack of features, poor usability, weak security, poor integration, manual workload, and scalability problems.	Critically evaluates limitations of previous studies: small datasets, weak validation, overfitting, bias, lack of explainability, limited real-world generalisation, and missing comparative experiments.
Research gap / project gap	Usually expressed as a system gap: existing systems do not provide a required feature, workflow, integration, user experience, or automation.	Usually expressed as a research gap: previous studies did not fully solve a data problem, compare models, explain predictions, handle bias, or validate results adequately.

5. Differences in Final Project Deliverables

Software Engineering Project Deliverables	Data Science Project Deliverables
Problem definition and requirement analysis	Research problem, research aim, objectives, questions, or hypotheses
Software requirement specification or equivalent documentation	Dataset source, dataset description, data dictionary, and data quality discussion
System architecture and design diagrams such as use case diagram, ERD, class diagram, sequence diagram, and deployment diagram	Exploratory data analysis with meaningful visualisations and interpretation
Database design and application development	Preprocessing, feature engineering, and model preparation
AI integration design, if AI is included	Training and comparison of suitable machine learning/deep learning/statistical models
User interface design and usability considerations	Model evaluation using appropriate metrics and validation methods
Testing evidence: unit, integration, system, user acceptance, performance, and security testing as relevant	Interpretation of results, limitations, bias, explainability, and ethical discussion
Working software system, source code, deployment evidence, and user manual	Prototype, dashboard, report generator, or system interface when required, supported by the research model

6. Suggested Literature Review Structures

6.1 BSc (Hons) Software Engineering - Final Development Project with AI Integration

1. 2.1 Introduction
2. 2.2 Background of the Problem Domain
3. 2.3 Review of Existing Systems and Similar Applications
4. 2.4 Requirement Engineering and User Needs
5. 2.5 Software Development Methodologies and SDLC Justification
6. 2.6 System Architecture, Design Patterns, and Database Design
7. 2.7 AI Integration Approaches and Intelligent System Features
8. 2.8 UI/UX, Accessibility, and Human-Computer Interaction
9. 2.9 Security, Privacy, Scalability, and Deployment Considerations
10. 2.10 Software Testing and Quality Assurance Approaches
11. 2.11 System Gap and Justification for the Proposed Solution
12. 2.12 Summary

6.2 BSc (Hons) Data Science - Final Research Development Project with AI Integration

13. 2.1 Introduction
14. 2.2 Background of the Research Area
15. 2.3 Review of Previous Research Studies
16. 2.4 Dataset Review and Data Quality Issues
17. 2.5 Data Preprocessing and Feature Engineering Techniques
18. 2.6 Machine Learning / Deep Learning / Statistical Models
19. 2.7 Model Training, Validation, and Hyperparameter Tuning
20. 2.8 Evaluation Metrics and Model Comparison Methods
21. 2.9 Explainable AI, Bias, Fairness, Ethics, and Privacy
22. 2.10 Critical Analysis of Existing Research
23. 2.11 Research Gap and Proposed Contribution
24. 2.12 Conceptual Framework and Summary

7. How AI Integration Should Be Discussed Differently

AI Feature	Software Engineering Literature Focus	Data Science Literature Focus
AI chatbot	Discuss chatbot requirements, conversation flow, UI integration, backend API/service integration, user authentication, logs, security, and usability testing.	Discuss NLP model or chatbot model selection, dataset or knowledge base, training/fine-tuning method, evaluation metrics, hallucination risk, accuracy of responses, and explainability.
Prediction module	Discuss how prediction is integrated into the system workflow and how users will receive the predicted output.	Discuss dataset, target variable, preprocessing, feature selection, model comparison, cross-validation, accuracy/precision/recall/F1 or regression metrics, and error analysis.
Recommendation engine	Discuss the recommendation feature as part of the system, database design, user journey, and performance within the application.	Discuss recommendation algorithms, user-item data, collaborative/content-based filtering, ranking metrics, cold-start issue, bias, and model evaluation.
Computer vision feature	Discuss image upload, system interface, storage, API/model integration, performance, and user feedback.	Discuss image dataset, preprocessing, augmentation, CNN/transfer learning architecture, training, confusion matrix, precision/recall, and generalisation limitations.

8. Evaluation Perspective for Supervisors and Examiners

When evaluating the final project, the examiner should not judge both degrees using the same emphasis. The expected academic contribution differs.

Evaluation Area	Software Engineering Emphasis	Data Science Emphasis
Problem definition	Clear user, organisational, or process problem leading to a software solution.	Clear research problem that can be investigated using data and models.
Literature review quality	Strong comparison of systems, technologies, design approaches, and development practices.	Strong critical comparison of studies, datasets, algorithms, metrics, and research gaps.
Methodology	Appropriate SDLC, requirement gathering, system design, implementation, and testing methodology.	Appropriate data science methodology such as CRISP-DM or similar research workflow, with experimental design.
Implementation	A complete and usable software system with proper architecture and code quality.	A reproducible data pipeline/model with meaningful experimental results; system interface is supportive rather than the only output.
Testing/evaluation	Software quality and user acceptance are very important.	Model performance, validation, comparison, and interpretation are very important.
Contribution	Practical system contribution and improved user workflow.	Research contribution through data, modelling, analysis, and evidence-based conclusions.

9. Writing Guidance for Students

- Do not write the literature review as a list of summaries. Compare and analyse studies or systems critically.
- Use recent and credible academic references. Prefer journal articles, conference papers, books, and official technical documentation.
- Use tables to compare previous studies, systems, algorithms, datasets, metrics, and limitations.
- End the literature review with a clearly written gap statement.
- Make sure the literature review directly supports the methodology chapter.
- When using AI integration, clearly state whether AI is a system feature or the core research method.
- For Software Engineering, connect the literature review to requirements, design, implementation, and testing.
- For Data Science, connect the literature review to dataset selection, preprocessing, modelling, evaluation, and interpretation.

10. Conclusion

In summary, a BSc (Hons) Software Engineering final project should primarily demonstrate the ability to engineer a complete software solution. Its literature review should therefore focus on existing systems, software design, system architecture, technology selection, AI integration, usability, security, and testing.

A BSc (Hons) Data Science final project should primarily demonstrate the ability to conduct a data-driven research investigation. Its literature review should therefore focus on previous research, datasets, preprocessing, algorithms, evaluation metrics, model comparison, research gaps, explainability, ethics, and evidence-based findings.

Key statement: Software Engineering asks, "Can the student build a reliable and usable system?" Data Science asks, "Can the student use data and models to produce valid, evaluated, and meaningful research findings?"